15

Re: Interpreting Physical Theories

> To understand a scientific theory, we need to see how the world could possibly be the way that the theory says it is. An interpretation tells us that. The answer is not unique, because the question is not the sort of question to call for a unique answer.
>
> (van Fraassen 1991, 336–7)

Chapter 1 conjured the image of "generalist" philosophers of science clashing like Olympian gods over such epic questions as whether our best scientific theories are true, while far beneath their lofty heights, "particularist" philosophers of physics toil like badgers on such matters as the proper understanding of Haag's theorem. Ruing the suggestion that generalist inquiries could and should proceed in isolation from particularist ones, that chapter promised a particularist excursion through QM_∞ which would reveal lessons generalists would be well-advised to take to heart. This concluding chapter undertakes to characterize some of those lessons. They concern the related issues of scientific realism, physical law, and the nature of theoretical virtue.

15.1 An anatomy of scandal

Consider the realist who believes our successful scientific theories are (approximately) true on the grounds that those theories have withstood decades of scrutiny; predicted a wide range of phenomena, including unexpected phenomena, to a high degree of accuracy; underwritten advances both technological and conceptual; contributed efficiently to diverse regions of scientific practice; and in general exhibited all or most of the tangled panoply of virtues realists are wont to take as constitutive of good science. I would contend that, over the course of human history, the theory which most thoroughly realizes these virtues is QM.[1] Yet, after nearly a century of foundational

[1] QM notably lacks one virtue that might appear on the list: answering to whatever yen for explanation predisposes people to be realists in the first place. Many of the contributions to Cushing and McMullin (1989) explore this theme.

scrutiny, it remains utterly unclear what someone who believed QM would believe. Roger Jones puts it well: "Physicists don't know what deep explanatory structure of the microworld to be realists about" (1991, 191). This, I think, is a scandal. But it's a scandal whose origins reward investigation. The present section endeavors to dig up possible roots of the scandal. Following sections argue that scandal-producing conditions are rampant in QM_∞.

Laboring in abstraction, generalists have constructed a schematic argument for scientific realism, which I will call "the Miracles Argument." The *locus classicus* is Putnam's "The Nature of Mathematical Truth":

> [T]he positive argument for scientific realism is that it is the only philosophy that doesn't make the success of science a miracle. That terms in mature scientific theories typically refer . . . , that the theories accepted in a mature science are typically approximately true, that the same term can refer to the same thing even when it occurs in different theories—these statements are viewed not as necessary truths but as part of the only scientific explanation of the success of science. (Putnam 1975, 73)

Here's one way to reconstruct the argument:

The Miracles Argument

1. Theory T is successful.
2. T's truth is the best explanation of this success.

$\therefore$ T is true.

The scandal arises when this Miracles Argument is de-schematized, by setting T equal to QM. Then its first premise is extremely well supported. But those who think they know how to believe the conclusion subscribe to interpretations of QM that mortify most realists. The Many Worlds Interpretation commits zany metaphysics; the Bohm theory implies the mere instrumental adequacy of a theory (the special theory of relativity) most realists are happier believing than they are believing QM; the Bare Theory implies we're almost always wrong about almost everything; and so on. Opining that "God is subtle but not malicious," Einstein probably meant that quantum theory was too deranged to be true of a universe subject to *benign* divine oversight. Many realists of my acquaintance join Einstein in hoping that QM turns out to be false.

Considered through the mists of lofty abstraction, the Miracles Argument has real intuitive pull. But when it's brought down to earth and concretized as an argument for realism about a particular, extraordinarily successful empirical theory, the content of its conclusion, as well as its persuasive force, is unclear. What's going on?

The first and most obvious moral to draw from the predicament is that particularist philosophers, wrestling with interpretations of particular physical theories, give content to positions in the realism debate, even positions that aren't realist. What a realist believes when she believes a theory T is an interpretation of T; the contents

of various antirealisms about T can be spelt out in terms of attitudes toward the manifold of interpretations T admits.[2] This obvious moral, as it stands, is consistent with the two-levels picture, according to which generalist debates may as well proceed in isolation from particularist concerns. The force of the moral, on the two-level picture, is that once the generalists have decided what attitudes we ought to have toward empirically successful theories in general, particularists will supply the interpretations of specific successful theories towards which we're meant to strike those attitudes.

But it is possible to draw from the scandal morals which are more leveling, morals which call the two-level picture into question. Here are some examples:

[Moral?: the disunity of virtues] The nature of the virtues constituting T's success, invoked in Premise 1 of the Miracles Argument, *isn't a matter for generalists to decide*.

According to this moral, in the Miracles Argument it's not just the "T" that's schematic but also the "successful." Prediction, explanation, strength, systematization, novel extendability, and so on, may be virtues of scientific theories, but they aren't virtues that admit a uniform explication whose details are independent of the circumstances activating those virtues. The disunity of virtue could derail a deschematized Miracles Argument because a virtue like "explanatory success" might look a lot more truth-conducive when considered as though unified and in abstraction than it looks in particular instances where actual theories express it.

[Moral?: the disintegration of virtue.] There may be no single interpretation of a particular theory T under which all the virtues cited in the first premise of the Miracles Argument accrue to it.

The moral evokes the possibility that a successful scientific theory T *underdetermines its own interpretation*, and does so in such a way that different, and often incompatible, interpretations account for different parts of T's total virtuosity. What derails the concretized Miracles Argument in this case is that no single interpretation possesses the totality of virtues cited in premise (1) as reasons to believe (given premise (2)) that T is true. Since to accept the conclusion of the concretized Miracles Argument is to believe *an interpretation of* T, the concretized Miracles Argument commits a fallacy of many terms.

Chapter 1 suggested that one motivation for cleaving to the ideal of pristine interpretation is to prevent the disintegration of virtue. If we're vigilant against the contamination of our interpretations by "adulterating" factors keyed to particular applications of the theory we're interpreting, we foreclose the possibility that the interpretation our theory receives depends on its circumstances of application—and with it the possibility

[2] Here's a spelling out I like: "There is more than one tenable interpretation of quantum mechanics. It is true that each adds something to the theory. These additions are 'empirically superfluous.' . . . Quantum theory is well understood. How can I say this when there are several mutually contradictory, perhaps equally plausible, interpretations in the field? Well, I assert it exactly because we do have interpretations that have so far stood the test of debate" (van Fraassen 1991, 336).

that as the empirical virtues a theory exhibits shift from application to application, the interpretation that secures those virtues on behalf of the theory shifts as well.

A variety of empirical success that most impresses many realists is the one for which Whewell coined the lovely term "consilience." A theory achieves consilience when it applies successfully to settings and problems distinct from those it was devised to address. Such novel successes strengthen the case for realism about a theory enjoying them only if it is *under the same interpretation* that the theory enjoys both its original successes and its novel ones. Here again, the ideal of pristine interpretation, the extremist commitment to interpret a theory in the same way no matter what conditions befall it, reinforces realism. So too does an extremist commitment to adopt a uniform interpretive strategy with respect to families of similar theories, so that the success of any of them redounds to the credit of all.

But commitment to pristine interpretation entails a risk: interpretations that abjure adulteration may thereby fail to make sense of the full (diverse, messy, conditioned?) range of empirical accomplishments that made us interested in the theory to begin with.

Ordinary QM can be understood to exemplify a moral even more dramatic:

[Moral?: the Decoupling of Truth and Virtue] *T* may have interpretations which *undermine* the Miracles Argument.

The infamous quantum measurement problem is a consequence of embracing the orthodox eigenstate–eigenvalue link along with the hypothesis that among the time evolutions described by the Schrödinger equation are those undertaken by composite object + apparatus systems in the course of measurement interactions. For it follows from these assumptions (along with further, apparently unobjectionable ones, about what kinds of interactions count as measurements[3]) that a measurement interaction will almost always leave its participating apparatus in a state that's *not* an eigenstate of the designated pointer observable, and so not a state, according to the eigenstate–eigenvalue link, in which the pointer observable possesses a determinate value. Bluntly put, it follows that most measurements don't have outcomes. But measurements have to have outcomes, if QM is to be empirically adequate. For those outcomes are the phenomena QM secures its empirical adequacy by saving. Without phenomena to save, QM isn't empirically adequate.

The orthodox response to this measurement problem invokes what textbooks tend to call "the Postulate of Measurement Collapse"—the supposition that during the course of a measurement interaction, unitary Schrödinger evolution is interrupted by a measurement collapse. Schrödinger evolution is deterministic, continuous, and reversible; measurement collapse is a discontinuous, irreversible, indeterministic change of the state of the object system to one of the eigenstates of the observable measured. This makes measurement collapse a miracle in Hume's sense, a violation of

[3] For more on these, see the exemplary Busch, Lahti, and Mittelstaedt (1996).

the law of nature encapsulated by the Schrödinger equation. Now recall the Miracles Argument, whose crux is that the success of false theories would be miraculous. We've just seen that if collapse-free QM, as interpreted by the eigenstate–eigenvalue link, is true, it takes a miracle for measurements to have outcomes *period* (much less ones that accord with quantum statistical algorithm). That interpretation severs the entailment, presupposed by the Miracles Argument (as well as by many of its critics), between a theory's truth and its empirical adequacy. If QM as interpreted by the eigenstate–eigenvalue link is true, then it's not empirically adequate.

In an effort to restore the connection between a theory's truth and its empirical adequacy, we might try interpreting QM by means of the eigenstate–eigenvalue link *plus* the postulate of measurement collapse. Alas, the theory so interpreted is either inconsistent (because it contains a pair of dynamical postulates that contradict one another) or ill-formed (because it lacks a criterion clearly demarcating situations governed by Schrödinger evolution from those stricken by measurement collapse). Either way, it lacks prominent virtues constitutive of the sort of empirical success evoked in Premise 1 of the Miracles Argument. Either way, we are back in the ambit of the second moral: that only under different and competing interpretations does a theory exhibit different virtues inclining us to belief. And, I suggest, similar fates plausibly await other candidate interpretations of QM: each of them strips the theory of some of the virtues realists would regard as truth-conducive. This is exactly why debate over how to interpret the theory continues nearly a century after its inception.

15.2 Realism sophisticated

Section 15.3 will argue that theories of QM_∞ underdetermine their own interpretation, and do so in such a way that different interpretations bear different parts of their overall explanatory burden. This is the disintegration of virtue. Section 15.3 will also argue that concrete empirical virtues exhibited by theories of QM_∞ under particular interpretations can look less cohesive and truth-conducive than their schematic counterparts. This is the disunity of virtue. Both the disunity and the disintegration of virtue challenge the Miracles Argument for scientific realism sketched in §15.1. But few would consider the realist position presented in that section to be state of the art. Thus this section discusses more sophisticated scientific realisms, under the headings of "global realism" and "Structural Realism." The discussions are cursory, and fail to characterize these more sophisticated positions in great detail or with real precision. Their aim is to suggest that the phenomena of virtue disunity and disintegration also attenuate the persuasive force of arguments promising to support these more sophisticated realisms.

15.2.1 The global argument

Sometimes "local" and "global" versions of the Miracles Argument are distinguished (see e.g. Ladyman 2002, 210–19). The argument of §15.1 is "local;" it presents the

empirical success of a *particular* scientific theory as grounds for a realist attitude toward *that* theory. By contrast, the "global" version of the Miracles Argument, of which Richard Boyd is a leading advocate, takes the overall success of the many-splendored juggernaut that is "the mature sciences" as grounds for a realist attitude toward those sciences as a corporate body.[4]

For Boyd, scientific realism is "an empirical hypothesis which is justified because it provides the best scientific explanation for various facts about the ways in which scientific methods are epistemically successful" (1985, 3). Boyd calls "instrumentally reliable" those theories which "make approximately true predictions about observable phenomena" (Boyd 1985, 4). He thinks there are instrumentally reliable theories. But—in contrast, perhaps, to the "local" realist of the previous section—he does not regard the mere existence of these theories as the central reason to espouse scientific realism. Boyd lends more significance to the fact that the mature sciences frame and condition a *methodology* that is instrumentally reliable insofar as it furnishes "a reliable guide to the acceptance of theories which are themselves instrumentally reliable" (Boyd 1985, 4). Scientists devise theories and protocols for their assessment; having carried out these protocols, they reach judgments about the instrumental reliability of the theories in question, which judgments are borne out by the deliverances of subsequent science.

For Boyd, the crucial observation is that this instrumentally reliable method for assessing the instrumental reliability of theories *relies on a background constituted by theories making up the mature sciences.* This theoretical background guides judgments about which experiments are well-designed, which data patterns are significant, which conjectures are sufficiently plausible to reward investigation, and so on. Suppose that the startling efficiency of the prevailing methodology of the mature sciences demands explanation. Global realism about the mature sciences meets the demand. Prevailing methodology, a methodology informed through and through by those sciences, is instrumentally reliable because the claims of those sciences are approximately true, their theoretical terms refer, and so on. In short, the method works because the theories guiding the method, and guiding it in myriad interlocking ways, are right.

Boyd's position is, of course, immensely more subtle and extensive than the one just sketched. Still, the position just sketched constitutes a sophistication of the two-step local realism presented in §15.1, and we can articulate on behalf of this more sophisticated global realism a more sophisticated account of the empirical virtues realists regard as evidence of truth. The virtues contributing to the instrumental reliability of the mature sciences include the capacity of theories collectively composing the mature sciences to serve both as springboards for further theory development and as frameworks for understanding theoretical and experimental interventions. In virtue of the first capacity, the methodology of the mature sciences generates hypotheses

[4] I am grateful to Kevin Coffey for both writings and conversations about Boyd's brand of realism.

whose instrumental reliability it is able to assess in virtue of the second capacity. These are exactly the capacities best explained, according to the global argument, by the hypothesis of scientific realism.

If the outcome of this global argument is to be something more specific, and controversial, than a posture of goodwill toward the mature sciences and their capacity for self-improvement, that outcome must incorporate some realist attitude toward *present* mature sciences. The content of a scientific realism supported by the global argument would presumably be furnished by something like an interpretation of the conjunction of the mature sciences (if a consistent one can be found), or by a conjunction of interpretations of the mature sciences (if a consistent one can be found).

Whatever the details of that content, the global argument for scientific realism is just as susceptible to the disintegration of virtue critique as is the simple-minded local realist argument laid out in §15.1. The myriad successes constituting the instrumental reliability of the mature sciences function in Boyd's argument as reasons to accept the hypothesis of scientific realism. But if there is no "interpretation of mature science" which equips it with all those myriad successes, a gulf opens up between what we're being asked to believe and the reasons we're being given to believe it. Put in more concrete terms, if QFT's contribution to the instrumental reliability of the mature sciences is so structured that different, and rival, interpretations of QFT are required to make sense of that contribution, then it's not at all clear *what* belief in QFT is licensed by the instrumental reliability of the mature sciences.

I am supposing this to be a problem because I am supposing that to adopt the realist hypothesis with respect to our present mature sciences is to believe (among other things) that QFT is approximately true. If the latter supposition is false, I suspect that "realism about the mature sciences" is a realism deferred until such time as the regnant mature sciences grow out of the interpretive and theoretical awkwardnesses besetting our present mature sciences. Addressing this deference, §15.6 argues that while it may not be incoherent, neither is it compulsory.

15.2.2 Structural Realism

Another realist sophistication is Structural Realism, injected into contemporary debates by John Worrall, who attributes the basic position to Poincaré. Worrall finds it peculiar that the two most intuitively compelling arguments encountered in the debate over scientific realism support competing positions: the Miracles Argument, already outlined, is an argument *for* scientific realism; the Pessimistic Meta-induction—the argument that the history of science, as a history of discredited theories, constitutes forceful inductive evidence that our present theories will also be rejected as false—is an argument *against* scientific realism. Worrall markets Structural Realism as "the best of both worlds": a position *both* the Pessimistic Meta-induction and the Miracles Argument support.

The heart of Worrall's account is a move to reduce the inductive base invoked by the Pessimistic Meta-induction to past scientific theories which achieved consilience

by enjoying novel or surprising successes beyond the domain of phenomena those theories were introduced to save. Regarding the theories in this reduced inductive base, Worrall observes that often some aspect of a "discredited" theory lives on in its successors. Take for instance Fresnel's wave theory of light, which reposes in the graveyard of rejected science, to which it was dispatched by the success of Maxwell theory. Nevertheless:

> There was an important element of continuity in the shift from Fresnel to Maxwell—and this was much more than a simple question of carrying over the successful empirical content into the new theory. At the same time it was rather less than a carrying over of the full theoretical content or full theoretical mechanism (even in "approximate" form) . . . There was continuity or accumulation in the shift, but the continuity is one of form or structure, not of content. (Worrall 1989, 117)

The Miracles Argument tells us that our successful scientific theories are getting something right. The Pessimistic Meta-induction alerts us that they're not getting everything right. The Structural Realist takes both lessons to heart by espousing a realism about the "form or structure" of our best theories, a realism which explains their empirical success; these forms or structures inoculate themselves against pessimistic inductions by "carrying over" to successor theories.

In the literature subsequent to Worrall, Structural Realism bifurcates into "epistemic" and "ontic" varieties. Roughly, the former holds that our best reasons for belief in scientific theories are reasons for belief in their structures; the latter is a thesis about physical reality to the effect that what there is is structural. So deep and roiling are these waters that I dare not wade far into them. But I will offer some observations from the safety of shore.

The broad-brush portrait of Structural Realism is undeniably attractive. But it doesn't tell us how to bring the position into sharp focus. Operating at the level of schematic abstraction, consider a successful scientific theory T. What I'll call the Content Question for Structural Realism is the question of what a Structural Realist believes about T. Structural Realism is a genuine position in the debate over scientific realism only if the Content Question has an informative answer. (I suspect that there are versions of ontic Structural Realism ("the world is structure") that demur on the Content Question. I don't mean to charge those versions with triviality. I mean rather to classify them as metaphysical theses, rather than positions in the scientific realism debate.)

I've already said that the Structural Realist about T believes in T's structure, but what this comes to depends on how two further questions are answered. First, what is a theory's structure? Second, what is it to believe in that structure? I'll focus here on the former. A patently inadequate metaphor for the "structure" on which Worrall keys is the "theoretical DNA" that survives through successive generations. The metaphor suggests the following answer to the Content question: "the Structural Realist believes in those structural elements of our current best theories which will be preserved in (all?)

their successors." Bracketing questions about how to understand the word "structure," this is a position consistent with Chapter 14's "Sounder Principle" of interpretation, which enjoined us to regard as significant only those features of physical theories that all future physics will continue to regard as significant. But like the Sounder Principle, this answer to the Content Question has a pronounced pragmatic shortcoming. Regarding what good Structural Realists are to believe *now*, it gives us no guidance.

An informative answer to the Content Question won't defer to future theories. One strategy for developing an informative answer will rest on principles that, given a successful theory T, identify "the structure of T." This strategy faces several challenges, rooted in the same possibilities animating the morals §15.1 draws about the crude local Miracles Argument. T's empirical successes are presumably myriad: otherwise the question of realism about T lacks urgency. Also presumably, the Structural Realist will take some variation of the Miracles Argument to warrant belief that T gets the structure of physical reality approximately right on the grounds that getting the structure of empirical reality approximately right explains T's empirical success. But it could be that different, and rival, takes on T's structure explain different instances or types of T's empirical success. It could also be that no single take on T's structure explains enough of T's empirical success for a structuralized Miracles Argument to be persuasive. The challenge to the Structural Realist is to find a principle for identifying T's structure that recoups "enough" of T's empirical success.

Now it might seem that the Structural Realist has a neat and promising solution to the challenges just posed. Those challenges evoke the possibility that a theory enjoys different empirical success under different (and rival) formulations and/or interpretations. The neat Structural Realist rejoinder is that such apparently rival interpretations are in fact "representations of the same structure" (Ladyman 2002, 420). The idea is that Structural Realism averts worries about the disintegration of virtue by identifying a common (structural) content shared by all virtue-supporting interpretations of the theory.

Unfortunately, it's hard to see what the neat answer comes to for theories of QM_∞. Candidates for the structure of a theory of QM_∞ include abstract C^* algebras, concrete von Neumann algebras, and instances of the latter which stand to instances of the former as universal enveloping von Neumann algebras. Cast in terms of Structural Realism, what I've tried to argue in the foregoing chapters (and what I'll try to underscore in the next section) is this: while structures of each of these types underlie *some* of the empirical successes of theories of QM_∞, the collection of significant successes enjoyed by theories of QM_∞ defy attribution to a single type of structure. Empirical successes mediated by the particle notion, and explanations relating the energy of quantum field states to that of their classical predecessors, require the structure of a privileged irreducible Fock space representation; such a structure stymies the physics of broken symmetry and of equilibrium at non-zero tempeatures. An abstract algebraic structure broad enough to accommodate a full slate of possible dynamics for QFT in curved spacetime is too broad to sustain mean field dynamics of the

sort so successful in the thermodynamic limit. The heuristically powerful insistence on Poincaré symmetry disables a full-blooded explanation of quantum coherence. And so on.

Chapter 14 gave a reason to take coalesced structures seriously, notwithstanding the role idealizations play in constituting those structures. The reason was the possibility that those structures might persist in future theories, a possibility lent some plausibility by the role played by renormalization group techniques, devised to accommodate those structures, in shaping physicists' QFTs. This case for the possible *interpretive significance* of coalesced structures is different from, and weaker than, a case for realism (structural or otherwise) about theories in which those structures appear. Plausibly significant interpretations of a theory T are those that enable T to discharge its duties and exercise its virtues *as a theory of physics*. I have been urging that arguments for realism about T fall flat if it is only under different, and competing, interpretations that T discharges different, and significant, duties, and exhibits different, and significant, virtues. Thus a case that an interpretation of T is significant isn't on its own an argument that one should be a realist about T by believing that interpretation.

15.3 QM_∞ and the morals

We are in a position to give a partial account of what it takes for a theory T to exhibit the virtues animating these more sophisticated realisms—virtues those realisms regard as grounds for belief in the mature sciences or their constituents. T is virtuous if it exhibits explanatory reach, particularly an explanatory reach extending to novel and surprising phenomena. T is virtuous if it stands in relations of intercalation and continuity with present and future theories. So standing constitutes T's contribution to the instrumental reliability of the mature sciences. And T's continuities with future science supply that aspect of T about which epistemic Structural Realists are realists. Let us add these virtues to those celebrated by the simple-minded local realist, paradigmatically, T's predictive and explanatory success. In a line, the Miracles Argument sophisticated is that T's virtuosity warrants belief in T, because T's approximate truth is the best explanation of its virtuosity.

Section §15.1 insinuated that such arguments can become less compelling when de-schematized: when actual scientific theories are cast in the role of T. Once the notion of empirical virtue is concretized and made applicable to a particular, successful theory, virtues that appear unproblematically truth-conducive when considered in the abstract become complicated in a way that attenuates the Miracles Argument and its kin. This section chronicles the attenuations visited upon such arguments by features of QM_∞ discussed here.

"Fecundity"—the capacity to breed successful successor theories—is a virtue. Indeed, it's a virtue central to the sophistications (global and structural) of realism related in §§15.2.1 and 15.2.2. Boyd takes the wherewithal to successfully and

efficiently *extend* science to be of the first importance. The "structure" about which (some) Structural Realists would be realists is what's preserved over theory change. But de-schematized, "fecundity" fractures into a variety of traits that needn't be co-instantiable. Take QFT in curved spacetime (see Chapter 10, especially §10.3). To attribute to this theory the virtue of fecundity is to equip it with content making it a fruitful breeding ground for quantum gravity. But different, and competing, interpretations, enable the theory to exercise this virtue. Insofar as semi-classical quantum gravity is cultivated for hints about future quantum theories of gravity, QFT states must be Hadamard (i.e. such as to yield an expectation value to the stress-energy observable). Thus interpretations limiting physically significant states to those satisfying the Hadamard condition inculcate the virtue of fecundity. But insofar as the Hawking effect is pressed into service as the closest thing there is to "data" constraining quantum gravity, non-Hadamard states must be allowed to participate, for they are among the states which articulate the phenomenology of Hawking radiation. In this respect, interpretations limiting physically significant states to those satisfying the Hadamard condition *inhibit* the virtue of fecundity. Fecundity is a virtue QFT in curved spacetime exhibits under different, and rival, interpretations.

Explanatory success is a virtue. Explanations framed by fundamental particle notions are supported by restricting physically significant states of the quantum field to the folium of the irreducible Fock space representation circumscribing the particle notion in question. But such an interpretive strategy leaves no conceptual space for different states of broken symmetry, and so hamstrings explanations (e.g. of phase transitions and critical phenomena) that might operate in that space. There is no single interpretive approach to QM_∞ that recoups, on behalf of the set of theories of QM_∞, the totality of their explanatory accomplishments. Hence, there is no single interpretation of QM_∞, belief in which is supported by that totality of explanatory success.

Chapter 12 argued that there are *individual* explanations offered by theories of QM_∞ no extremist approach to interpreting QM_∞ could secure. In particular, no extremist interpretation of QM_∞ could single-handedly enjoy the success of explaining phase structure. The explanation considered rested upon *disjoint* states *distinguished by* a macroscopic observable marking the difference between different phases. The Algebraic Imperialist and Universalist posit enough states to frame the explanation—but don't recognize the distinguishing observable as an observable. With a suitable choice of privileged folium, the Hilbert Space Conservative supports the distinguishing observable central to the explanation—but she doesn't recognize the disjoint states between which the observable distinguishes. The virtue of underwriting the explanation of phase structure is not a virtue any single pristine interpretation of QM_∞ exhibits on its own. No single pristine interpretation supports the whole explanation.

The foregoing serve as examples of individual virtues, regarded as truth-conducive by realists, breaking up on the interpretive shoals of theories of QM_∞—examples of what §15.1 labeled "the disunity of virtue." QM_∞ also illustrates what that section

called "the disintegration of virtue"—the tendency of sets of virtues, regarded by realists as collectively truth-conducive, to dissociate themselves, and accrue to a theory only under different and rival interpretations. Here are a few examples.

Consistency with other well-established theories, and capacity to contribute to instrumentally reliable methodology are virtues—virtues the global realist position in particular supposes to cleave together. Let's suppose STR is a well-established theory, consistency with which is a virtue for theories of QM_∞. Interpretations of those theories confining physical possibilities to representations satisfying Poincaré covariance exhibit this virtue (see Ch. 5). But if the Standard Model is reckoned to be a well-established theory, consistency with which is a virtue for theories of QM_∞, the plot thickens. Although the Standard Model is itself consistent with STR, interpretations of QM_∞ fostering assessments of the Standard Model's instrumental reliability may not be. A small part of that assessment relies on models of "soft photons" involved in scattering interactions confirming the Standard Model. But soft photon modeling is accomplished by coherent state representations violating Poincaré symmetry (see §10.4.5). Interpretations which make sense of a small part of free QFT's contribution to the instrumental reliability of the mature sciences fail what seemed to be the bedrock desideratum of consistency.

This example of virtue disintegration might prompt some to exclaim: "So much the worse for soft photons!" But even supposing we reinforce the virtue of consistency with STR by restricting physically admissible QFT states to those in the folia of Poincaré-invariant representations, the disintegration of virtue threatens. For explanatory power is another virtue. And, as §10.4.5 recounted, certain explanations of the phenomenon of coherence, a phenomenon exhibited by many states passing our test of consistency, make use of a phase observable. According to this explanation, the coherence of coherent states derives from their status as eigenstates of the phase operator: this status insulates them from phase fluctuations which would undo the correlations (constitutive of coherence) between values of the electric field at different spacetime points. This explanation succeeds only if there is such a thing as a phase observable. And there is, but only in representations violating the requirement of Poincaré symmetry, instituted to preserve the virtue of consistency with STR. That virtue and the virtue of explaining the phenomenon of optical coherence disintegrate: no interpretation underwriting one underwrites the other.

15.4 Law and possibility

I have suggested that, when it comes to theories of QM_∞, phenomena of virtue disunity and disintegration are rampant. This undermines arguments for scientific realism deschematized to be arguments for realism about theories of QM_∞. But the phenomena also have repercussions for how we think about physical law and physical possibility, repercussions which are the subject of this section.

In slogan form, my contention is that, when it comes to QM_∞ at least, there is a pragmatic dimension to theory articulation. What set of possible worlds we associate with a theory of QM_∞ can depend on what we'd like to do with that theory: what explanations, involving which magnitudes and guided by what laws, we aspire to; what phenomenological models we need to construct; what projects of theory development we'd like to sponsor.

To endorse such "pragmatization" of theoretical content might seem flaky. But it is kin to a hard-headed empiricist position John Earman articulates in his 1986 *Primer on Determinism*. The setting is the "best systems" account of physical law, the account Earman reckons to have the best chance of passing what he terms "The empiricist loyalty test" (1986, 85) that any two possible worlds that agree on all occurrent facts also agree on laws.[5] As formulated by David Lewis, the best systems account holds that "a contingent generalization is a law of nature if and only if it appears as theorem (or axiom) in each of the true deductive systems [i.e. deductively closed, axiomatizable sets of true sentences] that achieves a best combination of simplicity and strength" (1973, 73). Much more should be said about what constitute "strength," "simplicity," and their best balance. To make the point that's initially relevant here, I will suppose for now that these commodities are sufficiently well understood.

Let us say that world w is *nomically accessible* from a world w' if and only if all the laws of w' are true of w. It follows quite plausibly from the best systems account that "nomic accessibility" isn't a symmetric relation. For a vivid illustration of why, let world W be a simplified solar system, best systematized by Newton's laws, including the Law of Universal Gravitation. Let world W' be a world consisting of a single spherically symmetric mass. Now W's laws are true of W', making W' nomically accessible from W. But W's laws aren't axioms in the simplest strongest deductive system with respect to the totality of W' facts: deductive systems omitting W laws uninstantiated in W' will be just as strong but much simpler. Supposing that the best system for W' includes "For all x, x just sits there," the laws of W' aren't going to be true of W, which includes bodies that don't just sit there. So even though W' is nomically accessible from W, W *isn't* nomically accessible from W'. This is the asymmetry.

Here is Earman's commentary on it:

> For the empiricist, there are no irreducible modal facts. A world W is a world of non-modal facts. Uniquely associated with each such world is a set L_W of non-modal propositions true in W—the laws of W. To mark off the elements of this set we may prefix "it is physically necessary that," but that prefix is merely an honorific. Accessibility is a defined relation, not a metaphysical given: world W' is nomically accessible from W iff the L_W are all true in W'. There are then myriad subsets of physically possible worlds, each radiating outward from a logically possible world. No one of these collections is more powerful or potent than any other. (Earman 1986, 99)

[5] See van Fraassen (1989) and Earman (1993) for variant expressions of loyal empiricism.

Stripped of empiricist sermonizing, the upshot is that physical possibility is *naturalized* (cf. Earman 1995, 176–9), insofar as what's possible depends on *how the world is*: even *within* the space of worlds nomologically accessible from our own, what the laws are at each world varies with the facts of that world. Consider, by contrast, a view according to which the worlds nomologically accessible from our world are exactly the worlds which share the laws of our world (see e.g. Carroll 1994). This view leaves no scope for the laws of a nomologically accessible world to change with its facts.

Earman's picture has the feature that to each world w, there corresponds *one* circle of worlds nomically accessible *from it*. These are just the worlds of which w's laws are true (although they needn't be worlds for which w's laws are laws). In order to draw this picture, Earman needs to waive concerns about problematic ambiguities in the notion of best system.

Now, the coalescence account of physical content naturalizes physical possibility, for instance, when it suggests that local dynamical facts contour the space of possibilities accessible at the thermodynamic limit, shrinking it from the full set of states on the relevant canonical algebra $\mathfrak{A}$ to only those states on which W^* dynamics are well-defined. *Different* local dynamical facts can call for *different* shrinkages. What is the case shapes what might be.

But the coalescence account can also *pragmatize* possibility. The account allows that from the actual world W there may emanate many different sets of physically possible worlds, each indexed to a sense of "physically possible" *native to* W. But multiple, different, even rival senses of "physically possible" are native to W because W natives encounter a variety of scientific challenges that require different contourings of the space of physical possibility. The coalescence account allows that for the sake of modeling critical phenomena, the space of states possible for a system subject to such phenomena shrink from the full set of states on the relevant canonical algebra $\mathfrak{A}$ to only those states on which "phase" observables, abrupt changes in whose values mark such phenomena, are well-defined. Different contexts may call for different shrinkages. The coalescence account allows that for the sake of pursuing semi-classical quantum gravity, the space of possible configurations of a QFT on curved spacetime shrink from the full set of states on the relevant quasilocal algebra to only those states satisfying the Hadamard condition. The coalescence allows projects of axiomatic theory development which require representations to be Poincaré-invariant to be physical. But it also allows non-Poincaré-invariant infrared coherent states to guide the interpretation of scattering experiments.

When physical possibility is naturalized, sets of physically possible worlds are indexed with respect to the "anchor" world whose facts shape that set. When physical possibility is pragmatized, sets of possible worlds are indexed (or indexed *as well*) to circumstances of application within the "anchor" world. When physical possibility is pragmatized, there's a single way the world is, but (as gleaned by physics) there isn't a single set of ways it might be. Instead, there are many sets of ways it might be. Taking these sets seriously is part and parcel of commitment to a physical theory: to call

something an "electron" is to take on commitments regarding how it would behave in a variety of circumstances, and to offer explanations, evaluations, and further theory constructions in ways constrained by those commitments. But we take on different, internally cohesive collections of such commitments in different circumstances. Provided we can keep the circumstances calling for commitments straight, this doesn't make us incoherent. It makes us resourceful. And that suggests another way to conceive of empirical virtue.

15.5 Virtue reconceived

I don't think it's too controversial that theories have a modal dimension. What is controversial is how to analyze that dimension: as metaphysically basic, merely defined, or somewhere (if there is anywhere) in between. I am contending that "the" modal dimension of a theory is really manifold: at least some successful scientific theories, theories of QM_∞, are best understood not as determining a single set of possible worlds, sharply defined by principles of extremist interpretation, but as encompassing many such sets, at least some of which correspond to adulterated interpretations of limited but undeniable use. Here I want to suggest that this is no accident. I claim that it's a central theoretical virtue—it better enables a theory to discharge the many obligations it faces—to foster manifold spaces of physical possibility, spaces that are naturalized and pragmatized. Again, this isn't inconsistent. It's not saying theories contradict themselves with respect to how the world is. It's saying good theories are good theories because of a certain inbuilt flexibility about how the world might be.

The venerable realist thinks theories work so well because they're right: successful theories latch on to an unambiguous way the world is. I think there is an unambiguous way the world is. But I don't think latching on to it is the key to theoretical success. Unambiguously what it is, the world engages a physical theory in a variety of ways. It enhances a theory's capacity to emerge victorious from these engagements to embed its picture of how the world is in a variety of accounts of how the world might be. The account a theory offers of how the world might be is a window to its laws; they judge which logically possible worlds are also physically possible. A theory indulging in a variety of accounts of how the world might be is not a theory to which we can assign a unique and unambiguous set of laws. Even if laws are "best picture," because bestness is indexed to contexts, the right instrument for seeing the best picture is a kaleidoscope.

Theories explain, predict, model, and spawn new theories. These are all virtues. But what underlies or secures them isn't, or needn't be, truth. It can be ambiguity. This isn't to say anything goes. The ambiguity in question is severely *tempered* by our criteria for individuating theories, in the form of rooted physical relationships central to those theories. These criteria won't tolerate unlimited variation in the interpretations a theory receives. Those interpretations must be consistent with our sense of *what that*

theory is. But that sense rarely issues in a unique interpretation, and when it admits many interpretations, the manifold of interpretations functions as a theoretical resource.

This reconception of theoretical virtue extends a metaphor notorious from van Fraassen's *Scientific Image*:

> I claim that the success of current scientific theories is no miracle. It is not even surprising to the scientific (Darwinist) mind. For any scientific theory is born into a life of fierce competition, a jungle red in tooth and claw. Only the successful theories survive—the ones which *in fact* latched on to actual regularities in nature. (1980, 40)

According to van Fraassen's metaphor, our best current theories abound in empirical adequacy because only abundantly empirically adequate theories are likely to survive the peculiar selection pressures of modern science. Extending this evolutionary metaphor, I take fitness—the trait or traits of a theory that promotes its survival—to consist not only in empirical adequacy, narrowly understood as saving the phenomena, but as empirical success more broadly construed. A theory that underdetermines its own interpretation is like a healthy breeding population: it has a shot at enough diversity to (under some interpretation or another) meet the variety of demands its scientific environment places on it. Like survival, "empirical success" is a convoluted, chancy, and conditioned thing. Like genetic diversity, possibility pragmatized situates its possessor to respond successfully to the changing circumstances on which its survival depends.

15.6 Fundamental physics

Realists might move to dismiss the morals suggested in this chapter on several grounds. The morals are inspired by theories of QM_∞, and indeed by some of their less central applications. One grounds for dismissal attests, "Present quantum theory is bound to be abandoned in favor of some theory of quantum gravity that has a straightforward interpretation palatable to the realist." The testimony expresses a concrete version of the hope that "fundamental" physical theory will be free of those features of our current best theories which are problematic for realism. A variation on these grounds objects that many of the applications I've discussed aren't "fundamental." We should interpret theories only in light of their "fundamental" applications. In sum, the morals drawn are to be dismissed because their origins are dubious: they come from physics that isn't fundamental.

I should confess that I myself am not in the grip of the idea of Fundamental Physics. Why believe that there is any such thing? Why expect that there will ever be a *theory*—as opposed to a "way the world is"—fundamental in the relevant sense (whatever that sense is exactly)? What are philosophers of physics supposed to talk about in the meantime? But I will set these idiosyncratic hangups to one side.

More to the point, I prefer to understand issues of scientific realism to animate our engagement with current science. According to the Fundamentalist dismissal, the only

theory about which the question of realism arises in earnest is Fundamental Physics. The central difference between me and Fundamentalists lies not only in the attitudes we take to the theories ready to hand but also in the attitude we take toward a theory decidedly not ready to hand (and possibly inevitably present at hand): Fundamental Physics. Even if the very idea of Fundamental Physics is coherent, we have little inductive evidence that the physical theories humans develop and deploy will ever conform to the ideal of Fundamental Physics. I think this partly because scientific practice selects for theories whose reserves of ambiguity are empirical resources, and I suppose Fundamental Physics, whatever it is exactly, to harbor no reserves of ambiguity. So while I concede to Fundamentalists the possibility of Fundamental Physics, I think they owe me an argument making that possibility relevant to the attitude we ought to have about the theories we do have, and the theories we'll have after them, and the theories we'll have after *those* theories, and so on.

I haven't offered any proofs that future, even near future, science *won't* realize the possibility of Fundamental Physics. So I don't consider Fundamentalism irrational. It just takes faith in physics of a different order than the faith I've got.

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